2025 International Conference on Sustainability, Innovation & Technology (ICSIT)

YouTube Trending Video Analysis Using Natural Language Processing and Machine Learning

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2025 International Conference on Sustainability, Innovation & Technology (ICSIT) | 979-8-3315-3549-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICSIT65336.2025.11295060

***Abstract*— In today's digital world, platforms like YouTube generate a massive amount of user interaction data every single day. This paper takes a closer look at what makes certain YouTube videos trend by analyzing patterns behind their popularity. Using tools from Natural Language Processing (NLP) and Machine Learning (ML) in Python, the study examines video metadata, user sentiment, and key engagement metrics like the like-to-view and comment-to-view ratios. To understand how viewers feel about the videos, sentiment analysis is done using TextBlob. Topic modeling is carried out using Latent Dirichlet Allocation (LDA), which helps uncover common themes in video titles and the general emotional tone. For prediction, models like Linear Regression and XGBoost are used to model the relationship between engagement features and view counts. The results indicate that both video metadata and audience sentiment significantly contribute to a video's popularity. Furthermore the explanatory models—particularly Linear Regression and XGBoost—demonstrated strong performance in explaining the variance in view counts. These findings highlight the value of data-driven approaches in understanding content performance and offer a foundation for optimizing video strategies on dynamic platforms like YouTube.**

Keywords*— *YouTube, NLP, trending videos, sentiment analysis, topic modeling, machine learning

I. INTRODUCTION

YouTube stands out as a major digital platform for content sharing and user engagement. Understanding what drives a video to trend is not only a data science challenge but also a key concern for creators and businesses, given the vast volume of content uploaded daily. This study investigates the underlying patterns and factors associated with popular YouTube videos by applying ML and NLP techniques. The study first preprocesses a set of well-known American videos to extract crucial metadata, such as view counts, likes, dislikes, and comment statistics. NLP techniques are used to evaluate sentiment and reveal hidden topics in videos, while feature engineering is used to compute derived metrics like engagement rate and like-dislike ratio velocity. This study seeks to respond to fundamental queries like: Which trending videos are most associated with sentiment? What are the recurring themes in popular content? What are the most significant factors associated with high view counts in trending videos? What type of topics are more likely to be viral than the others? What themes or sentiments show more probability to be viral? The findings provide practical insights for data scientists, content producers who are interested in digital media analytics.

II. LITERATURE SURVEY

Various studies have implemented different types of ML techniques to predict virality of YouTube videos. Patil and Mishra [1] illustrated the efficacy of using XGBoost in optimizing popularity prediction, highlighting the strengths of using gradient boosting over basic models. Shah and Sinha [2] also used deep learning methods to identify intricate patterns in metadata, producing more accurate results than conventional methods. Rao and Jain [3] highlighted the predictive value of engagement measures like likes, dislikes, and comment numbers. Other studies, such as those by Silva and Fernando [4] and Caldera et al. [5], investigated classification techniques to identify trending content, showing the applicability of categorical models to early trend detection.

Some studies have pointed to the use of social and contextual cues in improving predictive accuracy. Singh and Kaur [6] proposed a hybrid model that integrates social-aware and content-aware features for optimal video delivery. In a connected vein, Bose and Gupta [7] used deep learning both for popularity forecasting and content filtering, mirroring the growing overlap between prediction and classification tasks. Davidson et al. [8] and Wu et al. [9] studied YouTube's recommendation networks and viewership patterns, providing insights into how algorithmic processes affect viewership.

Zhao et al. [10] included multimodal and text-based data and merged visual, audio, and metadata features for prediction. Tran and Lee [11], Putra and Muharram [12], and Wollmer et al. [13] employed comment or speech sentiment analysis to estimate audience emotion as an indicator of future popularity. Ensemble models, as introduced by Gupta et al. [14] and Irshad et al. [15], combined features like titles, tags, and metadata to improve the accuracy of prediction. Furthermore, research by Niture [16], Pillai et al. [17], and Wake et al. [18] applied supervised learning techniques for video classification and ranking of trending content,

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highlighting the significance of data preprocessing and feature selection. Trend-based research by Hasan et al. [19], Singh [20], and Meshram et al. [21] Ramalakshmi et al. [22] and Patel et al. [23] also provided insights into categorizing videos based on content types and predicting view counts per category. Few works, like Liu [24] and Prabha et al. [25], explored sentiment and content structure but lacked deeper topic modelling techniques.

Despite the variety of models explored, most existing work either leverages engagement metrics alone or relies on black-box deep learning techniques without interpretable feature engineering. Additionally, few studies explicitly incorporate topic modelling from titles or combine sentiment analysis with structured metadata in a ML pipeline. In contrast, the present study proposes a hybrid approach that combines sentiment scores, topic modeling (using LDA on video titles), engagement ratios, and metadata-driven features. By applying and comparing interpretable linear models (e.g. Lasso) and high-performance ensemble models (e.g., XGBoost, Gradient Boosting), this work offers a balanced framework that emphasizes both predictive power and model transparency. This positions our contribution as a unique and holistic addition to YouTube popularity prediction research, filling a notable gap in the literature.

III. Methodology

Explanatory modeling, feature engineering, NLP, and data preprocessing are all steps in this project's structured data science workflow. Every stage has been thoughtfully planned to enable explanatory analysis and extract valuable insights from data on popular YouTube videos.

A. Data Collection

The dataset comprised approximately 220,000 video records spanning the period from August 2020 to August 2023. Each entry included around 29 features, such as video titles, tags, publication and trending dates, likes, dislikes, and comment counts. The primary variable of interest, or target, was view count, which was log-transformed (log1p) to reduce skewness and stabilize variance for modelling purposes.

B. Data Preprocessing

The preprocessing phase involved handling missing values, removing irrelevant columns like descriptions, and parsing datetime fields for feature extraction Although extreme values were observed in features such as view count, likes, dislikes, and comment count, these were retained as they represent viral or highly engaging videos—an essential aspect of the prediction goal. Instead of removing them, log transformations were applied to reduce skewness and stabilize variance. The effects of these transformations are visualized in Fig. 1 and Fig. 2.

C. Natural Language Processing

NLP was applied primarily to the video titles and tags to extract meaningful textual features. First, the titles were This score was included as a feature in the prediction models to capture the emotional tone of the content. Additionally, topic modeling was conducted on the titles using LDA with a CountVectorizer. Each title was assigned a dominant topic (from 5 learned topics), and the resulting topic index was used as a categorical feature—later one-hot encoded for linear models.

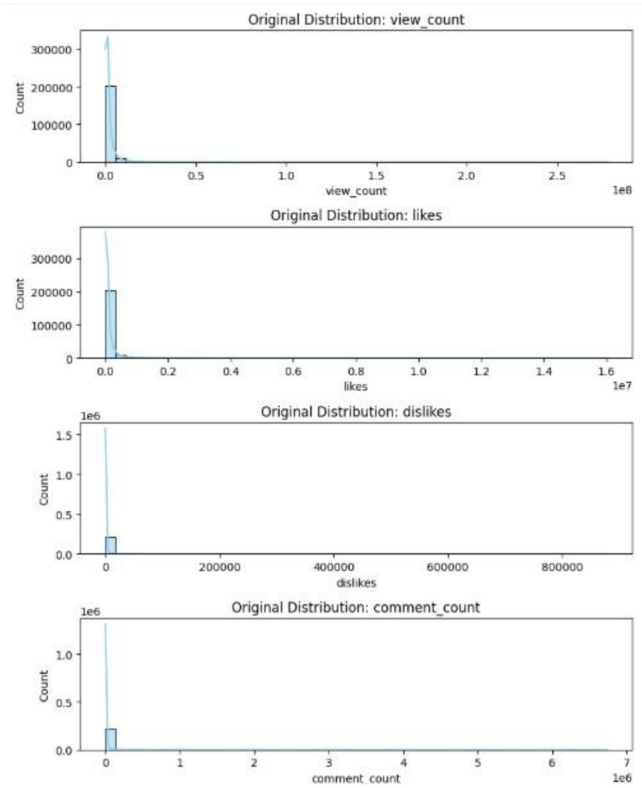

Fig. 1. Original Distribution of various parameters

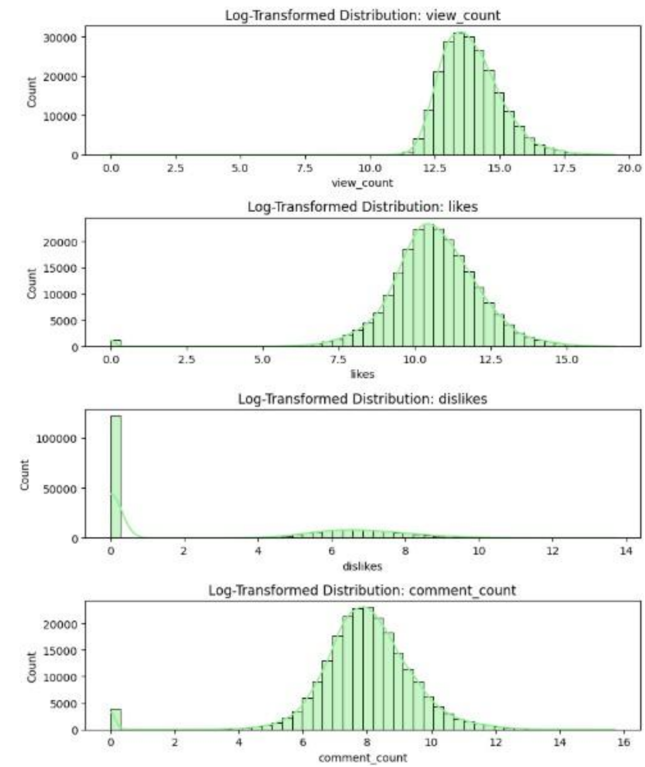

Fig. 2. Log transformed distribution of parameters

D. Feature Engineering

To enhance explanatory performance and extract deeper insights, several new features were engineered. Engagement metrics such as engagement rate, like dislike ratio, comment view ratio, and dislikes per comment were calculated to quantify audience interaction. Additionally, temporal features like days since publication, likes per day and comments per

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day were derived to capture the pace and timing of video engagement.

E. Popularity Modelling

To predict the popularity of YouTube videos, various ML models were implemented and evaluated. These included both linear and non-linear regressors. The linear models used were Linear Regression, and Lasso Regression, which are effective for baseline comparisons and interpretability. For capturing more complex patterns, tree-based ensemble models such as the Decision Tree Regressor, Gradient Boosting Regressor, and XGBoost Regressor were employed. Each model was evaluated using 5-fold cross-validation and tested on a held-out set, with metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Feature sets included log-transformed numerical metrics, engineered engagement and temporal features, sentiment polarity scores, and LDA-derived topic encodings.

F. Predictive Baseline Model

To establish a methodologically sound baseline for true prediction, a separate XGBoost model was trained using only pre-publication features (sentiment_score and topic). This approach avoids data leakage and provides a realistic measure of how well future popularity can be forecasted. The performance of this predictive baseline model resulted in a Test MAE of 0.8849 and a Test RMSE of 1.1467, achieving a Test R^2 Score of just 0.0357.

IV. Results and Discussion

The analysis yielded insightful patterns from both the NLP tasks and predictive modeling. This section summarizes key observations across sentiment analysis, topic modeling, correlation analysis, and regression performance.

A. Sentiment and topic trends

Sentiment analysis revealed that most trending video titles had neutral to mildly positive sentiment. Very few titles exhibited strong negative sentiment, indicating that uplifting or curiosity-driven content tends to trend more as seen in Fig.3. Using LDA, five major topics were discovered in video titles, corresponding to categories such as general vlogs, gaming, sports highlights, music videos, and movies and trailers, as demonstrated in Fig. 4. This confirms the thematic clustering of viral content. Emotionally engaging or curiosity-sparking titles are more likely to go viral. Titles featuring keywords like “official”, “trailer”, or “announcement” showed higher sentiment polarity and engagement as shown in Fig. 5 below.

B. Statistical and visual analysis

The correlation matrix shown in Fig. 6 indicated strong positive correlations between view count and likes, view suggests negative feedback plays a smaller role in trending status.

C. Model Performance

In this study, both linear and non-linear regression models were employed to model YouTube video popularity as presented in Table I. Linear models—Linear Regression, and Lasso—served as baseline models due to their simplicity and

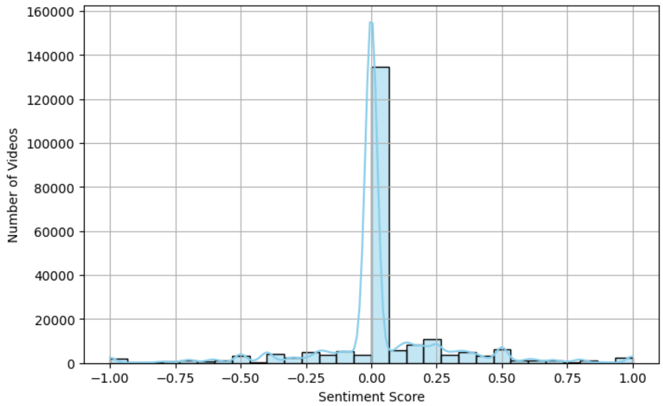

Fig. 3. Sentiment Scores in Video Titles

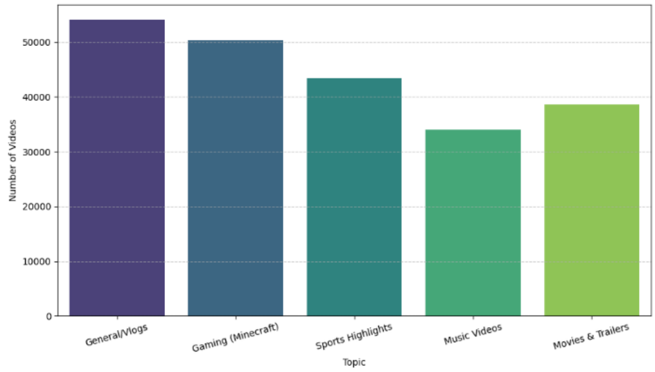

Fig. 4. Number of trending videos per LDA Topic

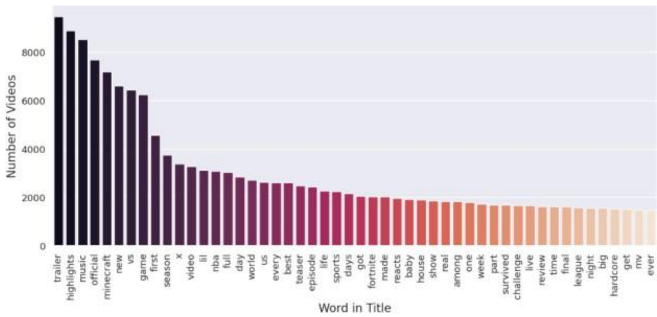

Fig. 5. Bar Chart showing frequently used words in video titles

interpretability. These models achieved moderate performance, with R^2 scores around 0.52–0.57 and MAE values above 0.54, indicating they could capture only basic linear relationships within the feature space.

In contrast, the non-linear models—Decision Tree, Gradient Boosting, and XGBoost—demonstrated significantly improved performance. Among them, XGBoost achieved the highest explanatory power, accounting for 96% of the variance in view counts and the lowest MAE of 0.0366, followed closely by Gradient Boosting and Decision Tree. Fig. 8 shows the scatterplot of actual vs predicted view count values. These models effectively captured complex, non-linear interactions. between features such as sentiment, topic, engagement metrics, and temporal variables. The superior performance of ensemble methods highlights their robustness and ability to generalize well, making them well-suited for modeling dynamic and multifactorial data like video popularity.

To further understand the drivers within the best-performing model, a feature importance analysis was

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conducted on the trained XGBoost regressor see Fig 7. The results reveal that log_likes is overwhelmingly the most significant feature, with an importance score that far surpasses all other variables combined. This indicates that the number of likes is the single most dominant factor in explaining a video's view count.

Furthermore, the analysis uncovers a clear hierarchy among engagement metrics, with engagement_rate and log_comment_count being secondarily important. A particularly insightful finding is the negligible importance of dislike-related features (log_dislikes, dislikes_per_comment). This suggests that a video's popularity is driven primarily by positive engagement, while negative feedback has a minimal role in explaining its total viewership. Features derived from NLP, such as sentiment_score and topic, also showed low importance, as their explanatory power is superseded by direct audience engagement metrics like likes.

TABLE I. MODEL PERFORMANCE COMPARISON

	CV MAE	Test MAE	Test R^2
Linear Regression	0.5412	0.5426	0.5697
Lasso Regression	0.5805	0.5825	0.5216
Decision Tree	0.0929	0.0933	0.9479
Gradient Boosting	0.0624	0.0638	0.9572
XGBoost	0.0363	0.0366	0.9609

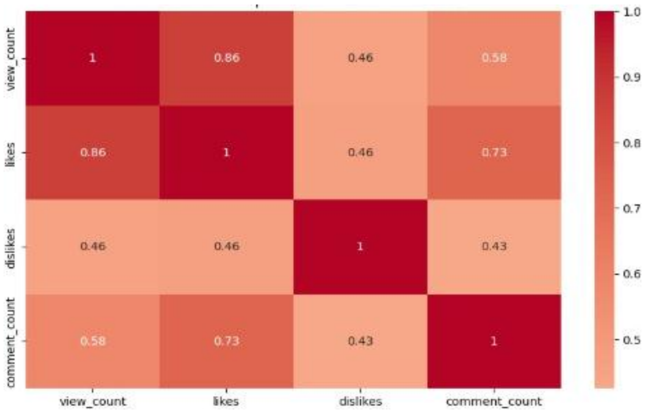

Fig. 6. Correlation matrix showing relation between numerical values

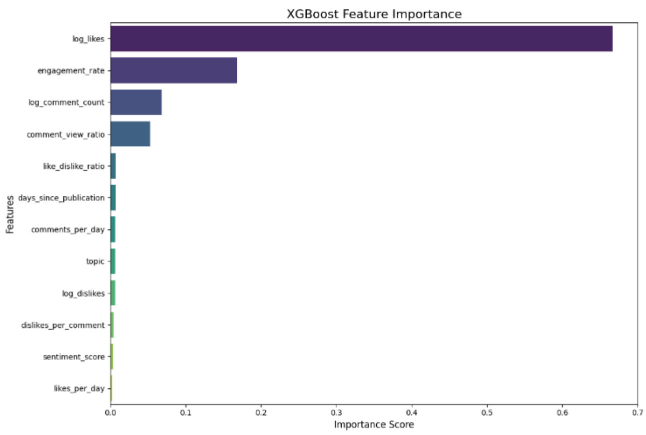

Fig. 7. Scatterplot of xgboost model showing actual vs predicted values

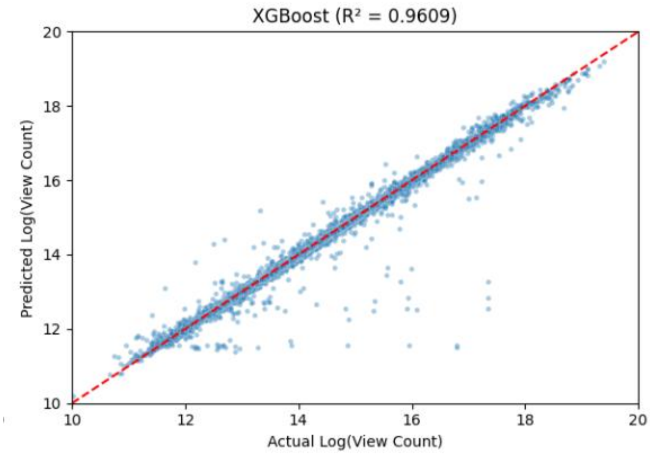

Fig. 8. Scatterplot of xgboost model showing actual vs predicted values

V. CONCLUSION AND FUTURE WORK

This project successfully demonstrates the use of machine learning and NLP to dissect the anatomy of popular YouTube videos. By applying sentiment analysis, topic modeling, and feature engineering, we confirmed that for trending videos, popularity is almost perfectly explained by post-publication engagement data. Our explanatory XGBoost model achieved a near-perfect R^2 of 0.9609, and a feature importance analysis revealed that a video's like count is the single most dominant factor, far outweighing metrics like comments or dislikes.

A key contribution of this research, however, is the stark contrast observed between explaining and predicting video popularity. To provide a methodologically sound comparison, a predictive baseline model was built using only pre-publication features. This model yielded an R^2 score of just 0.0357. This significant performance gap quantitatively demonstrates that while the success of a popular video is easy to explain in hindsight, forecasting it beforehand is an exceptionally challenging task. It underscores that the most valuable information for determining a video's performance lies in the audience interaction that occurs only after it is published.

Future work should focus on bridging this predictive gap. This can be done by engineering more sophisticated pre-publication features from video descriptions, tags, and thumbnails, or by analyzing creator-specific trends. Using deep learning models like BERT for a richer understanding of titles could also improve the predictive baseline, moving closer to a viable forecasting tool for content creators.

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